

Practical Work 7: Graph Neural Network Challenge

Initially, a simple model was applied mostly to better familiarize with the dataset and the task. Two simple GIN architectures, similar to the one used in the first exercise, were used to train the entire dataset. That is, one GIN to train the single task problems and another to train the three properties at the same time. The only thing that changes from one to another is that the GINMultiTask has twice as many nodes in each layer (so it can have a higher representative power) and it has a different output size.

In this architecture, only 3 GINConv layers with two-layer MLP aggregators are used, because graph networks don't need to be too deep. Additionally, Global additive pooling (global_add_pool) is used to aggregate node embeddings into a single graph-level representation for the prediction.

After training and computation of the official results on the test dataset, a few conclusions were taken to guide the modifications for the second version of the GIN. First, the training has taken more time than expected, the dataset is not large (not big data), but many hours were spend on each of the 4 training tasks. Therefore, hyperparameter experimentation has to be done wisely.

Second, after analyzing the results of the first version (table 1), we can see that there was no difference between the results for the single tasks and its multi counterpart. This is probably related to the wide variation on the properties scales, that is, a normalization preprocessing could be beneficial.

Version	prop1_mse	prop2_mse	prop3_mse	prop1_mse_multi	prop2_mse_multi	prop3_mse_multi
V1	0,8256	6,8565	0,0001	0,8256	6,8565	0,0001

Table 1 – MSE results obtained for the first submission.

For the second version of the GIN, the single task and multitask model distinction was abandoned since it had no effect in the first version, a unified model with versatile output dimension is used this time. Also, Batch normalization layers were implemented to try to stabilize training and better distinguish the properties in multi. In the concern of a three layers network being too deep, skip connections were used to mitigate the vanishing gradient problem.

Some changes were also made in the training process. It was made a normalization of the entire dataset to avoid the problem mentioned in the first session and a hyperparameter

search was implemented to find the architecture tailored for each task. A total of 36 combinations of Learning rate, batch size, number of layers and hidden dimension were evaluated for each of the four tasks.

That great amount of model trainings would take multiple days of training using the entire dataset, so for model selection only 20% of the dataset was used. We can see the selected hyperparameters in Table 2. We observe that a higher number of layers was chosen for all the models except for prop3, that might indicate an importance of long range interactions on those graphs.

Parameter	Prop1	Prop2	Prop3	Multi
Learning rate	0,001	0,01	0,001	0,01
Batch size	64	32	32	32
Nb of layers	3	3	1	3
Hidden dimension	128	64	128	128

Table 2 – Hyperparameters obtained after model selection.

Once the optimal hyperparameters had been identified for each task, four final models were trained using the full dataset and an increased number of epochs, under the assumption that the refined parameterization would lead to improved performance. However, an error occurred in the data normalization procedure: the target variable in the training set was inadvertently normalized. As a result, the models produced predictions in the range of 0 to 1, whereas the corresponding target values in the test set remained on their original scale. This mismatch led to the abnormally large mean squared error (MSE) values reported in Table 3.

Version	prop1_mse	prop2_mse	prop3_mse	prop1_mse_multi	prop2_mse_multi	prop3_mse_multi
V1	0,8256	6,8565	0,0001	0,8256	6,8565	0,0001
V2	7.990	6130.76	0.6951	7.990	78.06	6130.76

Table 3 – MSE results obtained for the first two submissions.

After identifying this issue in the challenge results worksheet, the models were retrained without applying normalization to the target variable. The corrected submissions were uploaded four days prior to the deadline in a third submission (CoelhoFernandes_GIN3.csv), however, the corresponding evaluation scores were not updated after that.

Additionally, once again the predictions for the individual tasks and the multi tasks were the same, it is an indication that the superior scale of prop2's value were not the issue. Therefore, the normalization did not solved that problem, but it might have helped with the overall score of the third submission, it is not possible to confirm.